

ARM-450 rev I.1 -- Modified (Craig) DH parameters

i	α_{i-1} (deg)	a_{i-1} (mm)	d_i (mm)	θ_i	θ_i from servo q_i	servo range	joint
1	0	0.000	90.000	θ_1	$\theta_1 = -q_1$	$q_1 \pm 90^\circ$	J1 base yaw
2	-90	0.000	0.000	θ_2 (home -90°)	$\theta_2 = -q_2 - 90^\circ$	$q_2 \pm 54^\circ$	J2 shoulder
3	0	119.000	0.000	θ_3 (home $+90^\circ$)	$\theta_3 = q_3 + 90^\circ$	$q_3 \pm 72^\circ$	J3 elbow
4	90	0.000	221.369	θ_4	$\theta_4 = -q_4$	$q_4 \pm 90^\circ$	J4 forearm roll
5	-90	0.000	0.000	θ_5	$\theta_5 = q_5$	$q_5 \pm 46.5^\circ$	J5 wrist pitch
6	90	0.000	0.000	θ_6	$\theta_6 = -q_6$	$q_6 \pm 90^\circ$	J6 tool roll

$${}^i_{i-1}T = Rot_x(\alpha_{i-1}) Trans_x(a_{i-1}) Rot_z(\theta_i) Trans_z(d_i) \quad {}^6_TT = Trans_z(d_T), \quad d_T = 88.089 \text{ mm}$$

$${}^0_TT = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T {}^6_TT \quad \text{home: } {}^0_TT = Trans_z(518.458)$$

Derived from the CAD servo axes and checked: every link transform has the exact Craig form; the frames match the model's to 1e-14 mm.
Spherical wrist (J4, J5, J6 meet at W, 430.369 mm above the base) -> closed-form IK.